

Initial contamination of chicken parts with Salmonella at retail and cross-contamination of cooked chicken with Salmonella from raw chicken during ...



The current study was undertaken to acquire data on contamination of chicken parts with Salmonella at retail and to acquire data on cross-contamination of cooked chicken with Salmonella from raw chicken during meal preparation. Whole raw chickens (n = 31) were obtained from local retail stores and cut into two wings, two breasts without skin or bones, two thighs, and two drumsticks. Data for cross-contamination were obtained by cutting up a sterile, cooked chicken breast with the same board and knife used to cut up the raw chicken. The board, knife, and latex gloves used by the food handler were not rinsed or washed before cutting up the sterile, cooked chicken breast, thus providing a worst-case scenario for cross-contamination. Standard curves for the concentration of Salmonella bacteria in 400 ml of buffered peptone water after 6 h of incubation of chicken parts as a function of the initial log number of Salmonella bacteria inoculated onto chicken parts were developed and used to enumerate Salmonella bacteria. Standard curves were not affected by the type of chicken part but did differ ($P < 0.05$) among the five isolates of Salmonella examined. Consequently, Salmonella bacteria were enumerated on naturally contaminated chicken parts using a standard curve developed with the serotype of Salmonella that was isolated from the original sample. The prevalence of contamination was 3 % (4 of 132), whereas the incidence of cross-contamination was 1.8 % (1 of 57). The positive chicken parts were a thigh from chicken 4, which contained 3 CFU of Salmonella enterica serotype Kentucky, and both wings, one thigh, and one cooked breast portion from chicken 15, which all contained 1 CFU of serotype 8,20:-:z(6). These results indicated that the poultry industry is providing consumers in the studied area with chicken that has a low prevalence and low number of Salmonella bacteria at retail and that has a low incidence and low level of cross-contamination of cooked chicken with Salmonella from raw chicken during meal preparation under a worst-case scenario.